in Figure 34 a. The procedure basically involves the isolation of genomic DNA, its digestion by restriction enzymes, separation by electrophoresis, and finally hybridization by incubating with cloned and labeled probes (Figure 12.34 b).

Based on the presence of restriction sites, DNA fragments of different lengths can be generated by using different restriction enzymes. In the Figure 12.34 b. two DNA molecules from two sources (A and B) are shown. In source A, a mutation has

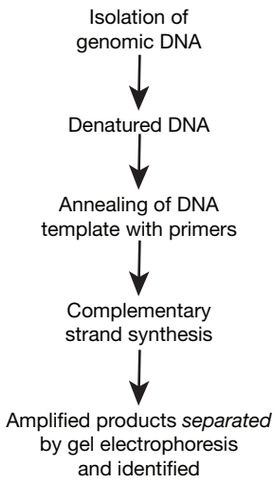


Figure 12.35: An outline of RAPD analysis as a molecular marker

Infobits

DNA Fingerprinting Or DNA Profiling:

DNA fingerprinting is the present day genetic detective in the practice of modern medical forensics. The underlying principles of DNA fingerprinting are briefly described.

The structure of each person's genome is unique. The only exception being monozygotic identical twins (twins developed from a single fertilized ovum). The unique nature of genome structure provides a good opportunity for the specific identification of an individual. The DNA fingerprint is an analysis of the nitrogenous base sequence in the DNA of an individual.

The original DNA fingerprinting technique was developed by Alec Jaffreys in 1985. Although the DNA fingerprinting is commonly used, a more general term DNA profiling is preferred. This is due to the fact that a wide range of tests can be carried out by DNA sequencing with improved technology.

Applications of DNA fingerprinting:

The amount of DNA required for DNA fingerprint is remarkably small. The minute quantities of DNA from blood strains, body fluids, hair fiber or skin fragments are enough. Polymerase chain reaction is used to amplify this DNA for use in fingerprinting. DNA profiling has wide range of applications – most of them related to medical forensics. Some important ones are listed below.

- Identification of criminals, rapists, thieves etc.
- Settlement of paternity disputes.
- Use in immigration test cases and disputes.

In general, the fingerprinting technique is carried out by collecting the DNA from a suspect (or a person in a paternity of immigration dispute) and matching it with that of a reference sample (from the victim of a crime, or a close relative in a civil case).

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